

# **Advanced Android**

## **From Code to Production**

This course takes you beyond the basics and turns you into a **well-rounded, production-ready Android developer**.

Whether you're an intermediate developer solidifying your foundations or an experienced engineer leveling up — this course covers the full spectrum of modern Android development, from architecture and language mastery to shipping and monetizing real-world apps.

Every topic is approached with a **practical, industry-focused mindset**.

This course is built around **Nami** — a unified AI chat app that brings all major LLMs into a single seamless experience.

Rather than isolated examples, every concept is applied directly to a **production-grade codebase** as you build Nami from the ground up.

By the end, you won't just have knowledge — you'll have a **fully functional, shippable application**.

ALL IN ONE

Switch between AI models seamlessly anytime

All AI models one place.

Access 30+ advanced AI models — GPT, Claude, Gemini, Llama and more — all unified in one fast...

The Nami app interface on a smartphone shows a list of AI models: Xai, Gemini, Grok, Claude, Anthropic, Open AI, and a '+more' button. Below the list is a text input field labeled 'Type a message...' with a 'GPT-5 pro' button. The background of the app interface is a dark, starry space with a nebula.

# Course Modules

1. Architecture & Design Patterns
2. Kotlin Deep Dive
3. Jetpack Compose & Accessibility
4. Networking, Data & Persistence
5. Performance, Optimization & Background Work
6. Security & Testing
7. Publishing, DevOps & Monetization

# Module 1

## Architecture & Design Patterns

Comparing Architectural Approaches, MVVM and State Management, Clean Architecture, Use Cases, Dependency Injection, Navigation Architecture, Error Handling Strategies, Modularization, Dynamic Feature Modules, Shared Business Logic in KMP.

# Module 2

## Kotlin Deep Dive

Null Safety & Type System, Scope Functions, Collections & Functional Programming, Extension Functions, Sealed Classes, Error Handling in Kotlin, Delegates, Generics, Inline Functions, Operator Overloading, Coroutines, Flow, DSL Building, Kotlin Contracts, Reflection & Annotations, expect/actual in KMP

# Module 3

## Jetpack Compose & Accessibility

Compose Fundamentals, Layouts, Lists & Lazy Layouts, State Management, Side Effects, Gestures & Touch Handling, Custom Drawing & Canvas, Theming, Material Design 3, Dark Mode, Animation, Responsive & Adaptive UI, Localization & i18n, Navigation, Interop with Views, Testing in Compose, Performance, TalkBack Support, Content Descriptions, Compose Semantics, Touch Targets, Font Scaling, Color Contrast, Accessibility Testing

# Module 4

## Networking, Data & Persistence

Retrofit, OkHttp, Kotlin Serialization, REST vs GraphQL, Caching Strategies, Interceptors, Error Handling, Firebase Firestore, Realtime Database, Authentication, Push Notifications, Remote Config, Cloud Functions, Room Database, DataStore, Encrypted Preferences, Offline-First Architecture, Database Migrations, Ktor & SQLDelight in KMP



# Module 5

## Performance, Optimization & Background Work

App Startup, Memory Management, Recomposition Optimization, Profiling Tools, Network Efficiency, APK Size Reduction, Baseline Profiles, Detecting Leaks with LeakCanary, WorkManager, Coroutines & Lifecycle, Foreground Services, AlarmManager, Battery Optimization, Doze Mode

# Module 6

## Security & Testing

Encrypted Storage, Biometric Authentication, SSL Pinning, ProGuard & R8, Secure Coding Practices, API Key Management, Unit Testing, UI Testing with Espresso, Compose Testing, Mocking with MockK, TDD, Integration Testing, Test Coverage, Shared Test Logic in KMP

# Module 7

## Publishing, DevOps & Monetization

CI/CD Pipelines, GitHub Actions, Firebase App Distribution, Play Store Release, Versioning, Signing Configs, Crash Reporting, Managing Build Variants & Flavors, Static Code Analysis, In-App Purchases, Subscriptions, Google Billing Library, AdMob Integration, A/B Testing with Firebase, Publishing Shared Modules in KMP